

LESSON 21 / 第二十一課

Phenology as a Clock: From Regional Seasons to the Bud

作為時鐘的物候：從區域季節到芽

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Phenology is a clock, not a calendar date. Start of season and end of season shift with temperature, photoperiod and, in some taxa, chilling and drought. Mixed evergreen and deciduous assemblages in the same city run different clocks. A flush that collides with a storm window is a load-and-carbon event, not a curiosity. Inspection envelopes that use only a date miss early and late years. The residual-risk sentence needs a time frame, a weather envelope and a phenological envelope together.

物候是時鐘，不是日曆日期。季初與季末隨溫度、光週期，以及在部分分類群中隨需冷與乾旱而移動。同一座城裡混生的常綠與落葉組合走不同時鐘。與風暴窗相撞的抽梢是荷載與碳事件，不是獵奇。只用日期的檢查包絡會錯過早年與晚年。剩餘風險句子需要時間框架、天氣包絡與物候包絡在一起。

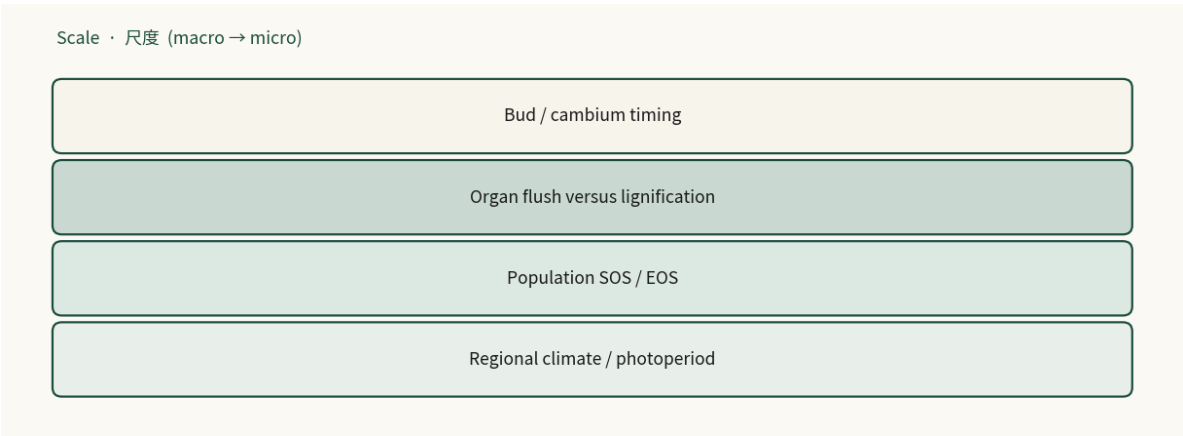


Figure 1. Nested scales from regional season to the bud.

圖1. 由區域季節到芽的嵌套尺度。



Figure 2. Flush and storm windows can collide.

圖2. 抽梢窗與風暴窗可以相撞。

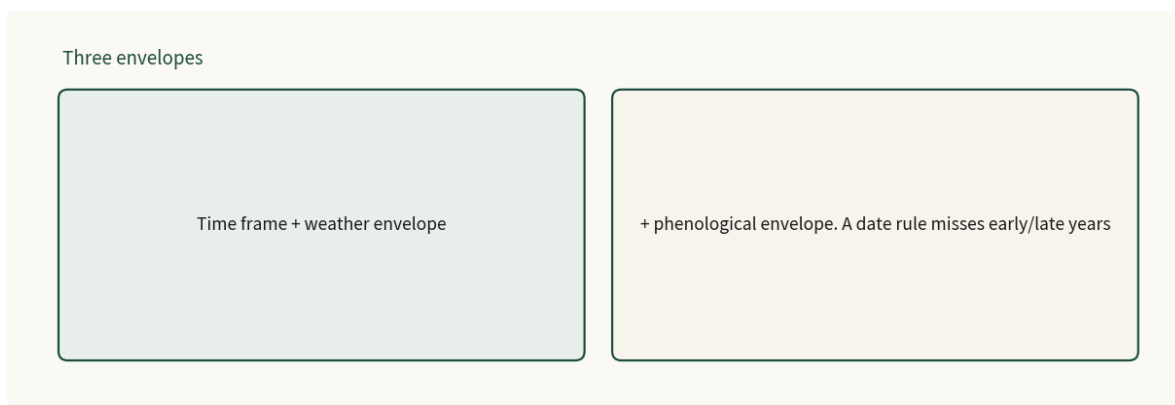


Figure 3. Risk sentences need a phenological envelope as well as weather.

圖3. 風險句子需要物候包絡，而不只是天氣。

1. Scope

1. 範圍

The object is published woody-plant phenology as a timing constraint on inspection, pruning and risk.

對象是作為檢查、修剪與風險之時間約束的已發表木本物候。

2. Clocks

2. 時鐘

SOS and EOS are population statistics derived from buds, leaves or remote sensing, not from a diary date [1,2]. Chilling, forcing, photoperiod and drought shift those statistics at different rates in different taxa [3]. Evergreen and deciduous guilds in the same street do not share one clock. A flush is a carbon and hydraulic spend; colliding with a storm window puts new, poorly lignified tissue into a high-drag envelope.

季初與季末是從芽、葉或遙感得出的族群統計，不是日記日期 [1,2]。需冷、促發、光週期與乾旱以不同速率在不同分類群中移動那些統計 [3]。同一條街上的常綠與落葉功能群不共享一個時鐘。抽梢是碳與水力開支；與風暴窗相撞，等於把新的、木質化不足的組織送進高阻力包絡。

3. Operational envelopes

3. 操作包絡

Pre-wet or pre-storm inspection that uses only a calendar date will be early or late in anomalous years [4]. Pruning timing is the same clock. The residual-risk sentence already has a time frame and a weather envelope; the phenological state of the part belongs in that sentence.

只用日曆日期的雨前或風暴前檢查，在異常年會偏早或偏晚 [4]。修剪時間是同一時鐘。剩餘風險句子已有時間框架與天氣包絡；部位的物候狀態屬於該句。

4. Limits

4. 限度

Temperate deciduous SOS models are thicker than tropical flush models. Transfer of a growing-degree-day threshold across biomes without a drought and photoperiod term is a guess.

溫帶落葉季初模型厚於熱帶抽梢模型。沒有乾旱與光週期項就把生長度日閾值跨生物群系轉移，是猜測。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. Phenology is treated here as:

1. 此處把物候視為：

- A. A fixed calendar date.
- A. 固定日曆日期。
- B. A clock.
- B. 一具時鐘。
- C. A t/R law.
- C. t/R 定律。
- D. A packet label.
- D. 包裝標籤。

2. SOS / EOS are:

2. 季初／季末是：

- A. Diary dates.
- A. 日記日期。
- B. Population statistics.
- B. 族群統計。
- C. Cable sizes.
- C. 鋼纜尺寸。
- D. Wall 4.
- D. 第 4 壁。

3. Evergreen and deciduous guilds in one street:

3. 同一條街上的常綠與落葉功能群：

- A. Always share one clock.
- A. 總共享一個時鐘。
- B. Do not share one clock.
- B. 不共享一個時鐘。
- C. Equal P50.
- C. 等於 P50。
- D. Equal 3-30-300.
- D. 等於 3-30-300。

4. Flush colliding with a storm window is:

4. 抽梢與風暴窗相撞是：

- A. A curiosity only.

- A. 只是獵奇。
- B. A load-and-carbon event.
- B. 荷載與碳事件。
- C. A register listing.
- C. 名錄登錄。
- D. An inoculant.
- D. 接種劑。

5. A date-only inspection envelope:

5. 只有日期的檢查包絡：

- A. Never misses.
- A. 永不錯過。
- B. Misses early and late years.
- B. 錯過早年與晚年。
- C. Equals QTRA.
- C. 等於 QTRA。
- D. Equals P88.
- D. 等於 P88。

6. The residual-risk sentence needs:

6. 剩餘風險句子需要：

- A. Only a date.
- A. 只有日期。
- B. Time + weather + phenological envelopes.
- B. 時間+天氣+物候包絡。
- C. Only paint.
- C. 只有漆。
- D. Only i-Tree.
- D. 只有 i-Tree。

7. Chilling and forcing:

7. 需冷與促發：

- A. Do not shift SOS.
- A. 不移動季初。
- B. Shift SOS at taxon-specific rates.
- B. 以分類群特定速率移動季初。
- C. Are lightning paths.
- C. 是雷電路徑。
- D. Are 10-20-30.
- D. 是 10-20-30。

8. Transfer of a GDD threshold across biomes without drought and photoperiod:

8. 沒有乾旱與光週期就跨生物群系轉移生長度日閾值：

- A. Is a measurement.
- A. 一項測量。
- B. Is a guess.
- B. 一項猜測。

C. Is Wall 2.

C. 第 2 壁。

D. Is a flare datum.

D. 幹基基準。

9. A flush is:

9. 抽梢是：

A. Free.

A. 免費。

B. A carbon and hydraulic spend.

B. 碳與水力開支。

C. A tomogram.

C. 斷層圖。

D. A soil nail.

D. 土釘。

10. Pruning timing uses:

10. 修剪時間使用：

A. A different physics.

A. 不同物理。

B. The same clock.

B. 同一時鐘。

C. Only Santamour.

C. 只有 Santamour。

D. Only Bond.

D. 只有 Bond。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

[1] Richardson A.D. et al. phenology network syntheses.

[2] Piao S. et al. remote-sensing SOS/EOS reviews.

[3] Korner C., Basler D. photoperiod and temperature control of spring phenology.

[4] Chuine I. process-based phenology models.

[5] Jim C.Y. mixed evergreen/deciduous urban phenology notes.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。